Interdisciplinary project



Course objective

- To train the students in **developing a full project**, considering the different phases of the **design** and **implementation** as well as to develop the ability of **team working** and **communication of the results**

- **Project based** learning approach. Apply the knowledge gained in the ICT area to the domains of health, intelligent transport systems, innovative building design, smart grids and geomatics/navigation

- Project are inspired by real-world problems of a “smart-society”

Characteristics of the course

- “Non conventional” with respect to other classes
  - Few hours of frontal teaching
  - The effort is spent in the group work
  - Methodology not usual in Politecnico, but widely used in other universities in the world

- Challenging opportunity but
  - Requires proactivity of the students
  - Independency
  - Ability to develop “problem solving”

Expected learning outcomes

- To learn problem-solving methodologies drawing on lessons from several disciplines and applying them in a very pr

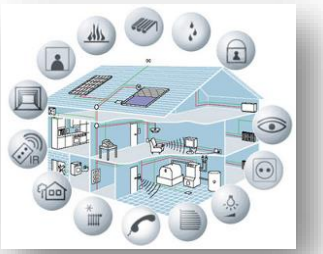


- Knowledge
  - Application of ICT technologies to other domains



- Abilities
  - Ability to work in a heterogeneous team
  - Ability to apply and select the proper ICT technology in a context different from the one in which the technology was originally deployed
  - Scientific communications





Organisation

- The course is formally split over 2 semesters
- Initial class hours are dedicated to:
  - Introduction of the different topics
  - Project management basics



- Group work (4-5 students)
- Supervision of a teacher/researcher
- About every 3-4 weeks each group prepares a progress presentation and provide updates on the progress of the work
  - progress
  - issues faced and solutions
  - few slides and presentation in English in front of the audience (teachers and classmates)
  - the presentation must be uploaded on the course webpage in the ”elaborati” folder



- A ICT4SS day may be scheduled at the end of the second semester for an open presentation of the results to general public

Appointment of the projects: procedure

- The teachers prepared a list of topics that will be published by next week. In order to have more information about the topics you should contact the reference persons for each topic. Most of the tutors will be in the class on Monday 28/09

- Students should express their preferences **using the Moodle application** on the portal before **03/10 eob**
- Each student will have to rank at least 3 topics in order of preference
- The topics will be allocated to maximize the satisfaction of the students according to their ranking of the topics, but any of the 3 topics could be allocated to the students
- By 05/10 the teams will be announced.
  - It will be possible to implement swaps of students between the groups. Students have to agree by themselves and communicate to prof. Dovis the swap to be implemented
- At least one topic per ICT area will be assigned.

Exam

- The exam is the delivery of the outputs of the project itself that will be presented and discussed in front of the teachers committee
- The group discusses the project **all together**

- A report on the project has to be delivered **at least 1 week before the exam** to the reference teachers and uploaded on the portal

- The Final Review (FR) of the project acts then as the exam of the course.
- The date of the FR can be arranged during the winter exam session or during the second semester, according to the planning of the project

Exam

- The exam grade is based for 1/3 on the quality of the progress presentations and the final presentation, and for 2/3 on the final oral exam during the Final Review and discussion of the final report (delivered 1week before the FR).
- The oral exam is a discussion on the whole project that has been developed and it will start from the delivered report, in order to test:
  - their understanding of the developed work of each student of the team
  - their acquired ability for the problem solving
  - the acquired knowledge of the ICT topics necessary for the project development
  - the actual results achieved
- Each team will be asked to perform a **self evaluation** of the team work and of each individual contributions through peer-review. The self evaluation is used to weight the grade of the project for each individual team member
- The interim presentation must show the ability of the team to manage the project, the organization of the team work, and a proper delivery mode (**correct technical language, clarity of presentation, ability to focus on the core ideas**).

Self-evaluation

- Before the final presentations each member of the team will be asked to perform an anonymous peer-review of the work of the other members of the team according to two criteria:
- 1- Technical contribution with respect to the expected tasks
- 2- Commitment in developing the project
- **The scores are anonymous for the other members of the team and for the teacher**, and have to be briefly motivated.

- The score achieved by each member of the team will be used to weight the overall grade of the project for the individual final grade.

Miscellanea

- The course takes 2 semesters but the project can be finished earlier
  - in particular for the students expecting to leave for thesis/stages during the second semester
  - the whole group must agree in finishing in advance

- The project **is not the thesis**! However, thesis topics might stem from the project itself and be further deployed as full thesis (with additional innovative contributions)

- In case issues arise within a group they have to be reported to prof. Dovis before the exam date

Miscellanea

- A Microsoft Teams channel will be created for the communications within the teams and with the teachers

- It should be used as the **preferred communication** mean for the exchange of information and material